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Authors	Kene Nwosu, Joy Vaz, Sabina Rodriguez Velásquez, Sara Botero Mesa, Olivia Keiser

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R Basics and Beyond: Outcomes of a Cohort-Based Online Programming Course for Health Professionals

Kene David Nwosu[1,2], Joy Venkatachalam-Vaz[2], Sabina Rodriguez Velásquez[1,2], Sara Botero Mesa[1,2], Olivia Keiser[1,2]

[1] Institute of Global Health, University of Geneva, Geneva, Switzerland
[2] The GRAPH Network, Geneva, Switzerland

Corresponding author:
Kene David Nwosu
Institute of Global Health, University of Geneva
Chemin des Mines 9, 1202 Geneva, Switzerland
kenechukwu.nwosu@unige.ch

ORCiDs:
Kene David Nwosu: 0000-0002-3245-4109
Joy Venkatachalam-Vaz: 0000-0001-8970-5805
Sabina Rodriguez Velásquez: 0000-0002-2758-7185
Sara Botero Mesa: 0000-0002-2943-3120
Olivia Keiser: 0000-0001-8191-2789

Keywords: Online learning; R programming; health data science; course completion; student experience; MOOCs

Abstract

Online education and open-source programming languages like R are expanding access to data science training, but most online offerings are self-paced and suffer from low completion rates, particularly for code-based content. Cohort-based online courses aim to combine the flexibility of online learning with the structure of guided programs, but evidence on their outcomes for data science training remains limited. We analyzed administrative roster data and post-course survey responses from 458 students enrolled across 9 cohorts of an online R course for health professionals (R Basics and Beyond) between 2023 and 2025, which used a flipped classroom model combining self-paced pre-recorded video lessons with weekly live workshops. The overall course completion rate was 55.9%. Among 206 post-course survey respondents, satisfaction was high (96% giving a recommendation score of 9-10 out of 10), with pre-recorded videos and live workshop facilitation rated highest. Time constraints and competing work priorities were the most frequently cited challenges. These findings demonstrate the feasibility of cohort-based online R training for health professionals while highlighting the need for greater scheduling flexibility.

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Public Interest Summary

The GRAPH Network is an international collaboration based at the University of Geneva's Institute of Global Health that develops training programs and analytic support to strengthen data science capacity in the global health workforce, with a particular focus on low- and middle-income countries. Our flagship online training program, the R Basics and Beyond bootcamp, was originally launched as part of the WHO African Regional Office's COVID-19 response and now teaches R programming and data analysis to health researchers and practitioners worldwide through a cohort-based, flipped-classroom format. The research reported in this paper is the first systematic evaluation of educational outcomes across multiple cohorts of this program, examining enrollment, course completion, participant satisfaction, and qualitative feedback. The findings inform our ongoing efforts to refine the cohort-based online format and contribute to wider work on scalable, accessible data science training for health professionals.

Introduction

As health information is increasingly digitized in electronic health records, outbreak surveillance systems, and routine reporting platforms, growing amounts of data are available at scale for clinical and public health decision-making. The ability to analyse these data is thus becoming a core competency for health professionals. Despite broad recognition of this need, surveys reveal significant skill gaps, with much health data remaining underutilized due to limited analytical capacity in health institutions[1]. This problem is particularly acute in low- and middle-income countries (LMICs) where training opportunities and analytic infrastructure may be more limited[2].

A range of approaches have been used to strengthen data skills in the health workforce. Short in-person workshops are a popular format, delivering intensive instruction with hands-on practice and real-time feedback[3]. Such workshops have long been favored by international health organizations for their ability to build competencies rapidly in targeted groups. However, these programs can be difficult to scale. As the WHO African Regional Office has noted, face-to-face training can only reach a limited number of professionals due to "costs, geographical limitations, and the time required to organize and conduct such events"[4].

Massive open online courses (MOOCs), which gained prominence in the early 2010s, offered a potential solution for delivering professional education at scale[5]. These courses are open to unlimited participants at minimal or no cost, allowing anyone with an internet connection to access educational content from world-class institutions. However, MOOCs suffer from a major limitation: completion rates are notoriously low, often single-digit percentages[6,7]. These self-paced courses lack the accountability, instructor interaction, and peer support that help learners stay on track[8,9]. Without these structures, many learners struggle to regulate their own learning, leading to disengagement and dropout[10].

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These challenges may be particularly acute for code-based data analysis training. Although health professionals have traditionally learned to work with data using point-and-click software such as SPSS, Stata, and Epi Info, the R programming language has become an increasingly popular alternative due to its flexibility, reproducibility, and zero licensing cost[11,12]. However, R has a steep learning curve, particularly for users without prior programming experience. Learning to code requires not only understanding syntax but also developing problem-solving and debugging skills—competencies that are difficult to acquire without guidance. Neither dominant training model is ideally suited to this challenge: self-paced online formats may leave learners stranded when they encounter errors and programming difficulty, while intensive in-person workshops may not provide sufficient time for the sustained practice required to build programming fluency.

Cohort-based online courses represent a potential middle ground between these approaches. These programs typically use a flipped classroom model, combining asynchronous content, such as pre-recorded videos and readings, with scheduled live sessions[13]. The asynchronous component allows learners to engage with material at their own pace while reducing total contact hours and delivery costs. The live sessions provide accountability, peer interaction, and direct instructor support—elements often lacking in purely self-paced courses[14]. Such guidance is especially valuable when teaching programming, where learners frequently encounter errors that are difficult to resolve independently. Notably, the normalization of videoconferencing during the COVID-19 pandemic has created new opportunities to deliver such programs at scale[15].

The GRAPH Network (Global Research and Analyses for Public Health) was established in 2020 as part of a collaboration to support the WHO African Regional Office's COVID-19 response[16]. One of its flagship training programs is the R Basics and Beyond (RBB) bootcamp, an online 12-week part-time course that introduces R programming and data analysis to researchers and practitioners in health-related fields. Delivered in partnership with the University of Geneva's Infectious Diseases and Mathematical Modelling unit, the bootcamp uses the cohort-based flipped classroom model, combining self-paced pre-recorded video lessons with weekly live workshops featuring collaborative problem-solving in small groups.

This study evaluated the educational outcomes of the RBB program across multiple cohorts from 2023 to 2025. Using administrative roster data and post-course survey responses, we assessed enrollment and completion rates, described participant characteristics, and analyzed open-ended feedback on course highlights and challenges using thematic coding. Our findings aim to inform the design of scalable, accessible data science training for health professionals worldwide.

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Materials and Methods

Study Setting

This retrospective cohort study analyzed enrollment, completion, and post-course survey data from the R Basics and Beyond (RBB) bootcamp, a 12-week online course designed to introduce R programming to health professionals.

The course used a flipped classroom model. Each week, students completed 2-3 hours of self-paced prework consisting of pre-recorded video lessons, PDF/HTML handouts, theory quizzes, and practical data-based quizzes. This was followed by a 2-hour live workshop held via videoconferencing software.

Workshops typically began with a short content review, followed by approximately 1 hour of small-group work (2-4 students) in breakout rooms where students collaborated on a coding assignment. Students were encouraged to share screens and discuss solutions with peers while working, but they submitted assignments individually. Workshops concluded with student presentations, instructor explanations, and closing discussion. One to three instructors were present for each workshop, with a target student-to-instructor ratio of 15:1.

Course content covered data manipulation with tidyverse, data visualization with ggplot2, and basic descriptive statistics, using health-relevant datasets throughout. Over the 12 weeks, students completed 10 workshop assignments, with three to five receiving detailed graded feedback and the remainder checked briefly. Feedback was typically returned within 1-2 weeks.

Students enrolled either individually or through institutional sponsorship. Course fees varied, with partial or full financial aid available to students who demonstrated need; funding categories and amounts were calculated and are reported in the Results section. All enrolled students retain indefinite access to course materials after enrollment.

Study Population

We included all students enrolled in RBB cohorts from May 2023 (RBB 2023 Q2) through April 2025 (RBB 2025 Q2), comprising nine completed cohorts.

Data Sources

Two primary data sources were used:

1. **Course rosters**: Administrative records containing enrollment status, graduation status, and country of residence for all students across all cohorts.
2. **Post-course survey**: An online survey administered via Google Forms 2-6 weeks after the final project submission deadline. The survey included a consent question asking respondents for permission to use their responses for research purposes.

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Measures from Course Rosters

Course completion was obtained from the course rosters. Course completion ("graduation") required achieving a passing grade of 80% based on: online quizzes (10%), graded workshop assignments (40%), workshop attendance and participation (10%, requiring attendance at a minimum of seven workshops), and a final capstone project (40%). For the final project, students independently imported, cleaned, visualized, and interpreted a self-selected real-world dataset using R Markdown. Projects were graded against a rubric requiring reproducible code, use of at least four dplyr data wrangling functions and at least four ggplot2 figures with interpretation, with output in HTML, PDF, or Word format.

Roster data also included country of residence, student status (yes/no), employment status (yes/no), and funding category (self-funded, partial scholarship, or full scholarship/institutional coverage).

Measures from Post-Course Survey

Demographic variables collected via survey included gender (man, woman), age group (21-30, 31-40, 41-50, above 50 years), highest education level (Bachelor's or less, Master's/Postgraduate, Doctorate/Medical degree), region of residence (Africa, Americas, Asia, Europe), financial aid status (paid full fee, received financial aid, institution paid), and whether English was the respondent's native language (yes/no). Prior experience was assessed for both programming languages and statistics/data science (categorized as none vs. some/significant experience).

Overall satisfaction was assessed using a likelihood-to-recommend score, derived from the question "On a scale of 1-10, how likely are you to recommend this bootcamp to a friend or colleague in your field?"

Four open-ended questions assessed: (1) how the course impacted professional or academic work, (2) course highlights, (3) challenges faced during the course, and (4) recommendations for improvement.

Statistical Analysis

Descriptive statistics were calculated for all variables. Categorical variables were summarized as frequencies and percentages. Continuous variables (course component ratings, likelihood-to-recommend scores) were summarized as means with standard deviations. Geographic distribution was mapped using country of residence data from enrollment records. Completion rates were calculated overall and by cohort.

To explore potential predictors of completion, logistic regression was used to estimate odds ratios (ORs) of completion with 95% confidence intervals for student status, employment status, funding type, and region of residence (Africa, Americas, Asia, or Europe), using roster data from all enrolled students. To address missing data in predictor variables, multiple imputation by chained equations (MICE) was performed using predictive mean matching with 10 imputations. The imputation model included all predictor variables, cohort, and the outcome variable (course completion). Both

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univariable and multivariable models were fitted on each imputed dataset and results were pooled using Rubin's rules; multivariable models were mutually adjusted for all predictors and additionally included cohort as a fixed effect. A complete case analysis was also conducted as a sensitivity analysis.

To assess potential response bias in survey data, likelihood-to-recommend scores and mean course component ratings were compared between survey respondents who reported completing the course and those who did not. All analyses were performed using R statistical software (version 4.2.2).

Qualitative Analysis

Open-ended survey responses were coded using a predefined codebook developed through iterative review of a subset of responses. Each response could receive multiple codes where applicable.

Coding was performed with large language model (LLM) assistance: responses were processed using Claude Sonnet 4 (Anthropic, accessed via OpenRouter), with a randomly selected sample of 20 responses (~10% of all responses) per question manually reviewed by the authors to ensure the LLM's interpretation aligned with the coding framework. Discrepancies were minor and resolved by the authors.

Code frequencies were calculated as the percentage of respondents whose response contained each code, and representative quotes were selected to illustrate key themes.

Ethical Considerations

This was a non-interventional study of anonymous responses to a course evaluation survey, supplemented by routinely collected administrative records. Survey participation was voluntary, and the survey included a consent question asking respondents for permission to use their responses for research purposes; only responses with explicit consent were included in this analysis. Because the study used only anonymous secondary data with documented participant consent and involved no intervention or contact beyond standard course evaluation, formal ethics committee review was not required.

Results

Enrollment, Funding, and Course Completion

Overall enrollment patterns and course completion were examined across all cohorts. A total of 458 students were enrolled across 9 RBB cohorts between 2023 and 2025, representing learners from 98 countries worldwide (Figure 1A). The overall course completion rate was 55.9% (256 students; Figure 1B-C). Completion rates varied by cohort, ranging from 45.2% (Jan-Mar '25; n=19 of 42) to 70.5% (Jul-Sep '23; n=43 of 61).

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Scholarship and funding support was substantial: 24.7% (n=113) of students received full scholarships or institutional fee coverage and 39.1% (n=179) received partial scholarships, while 36.2% (n=166) paid the full fee. Among students who made any payment, the median amount paid was $285 USD (IQR: $287).

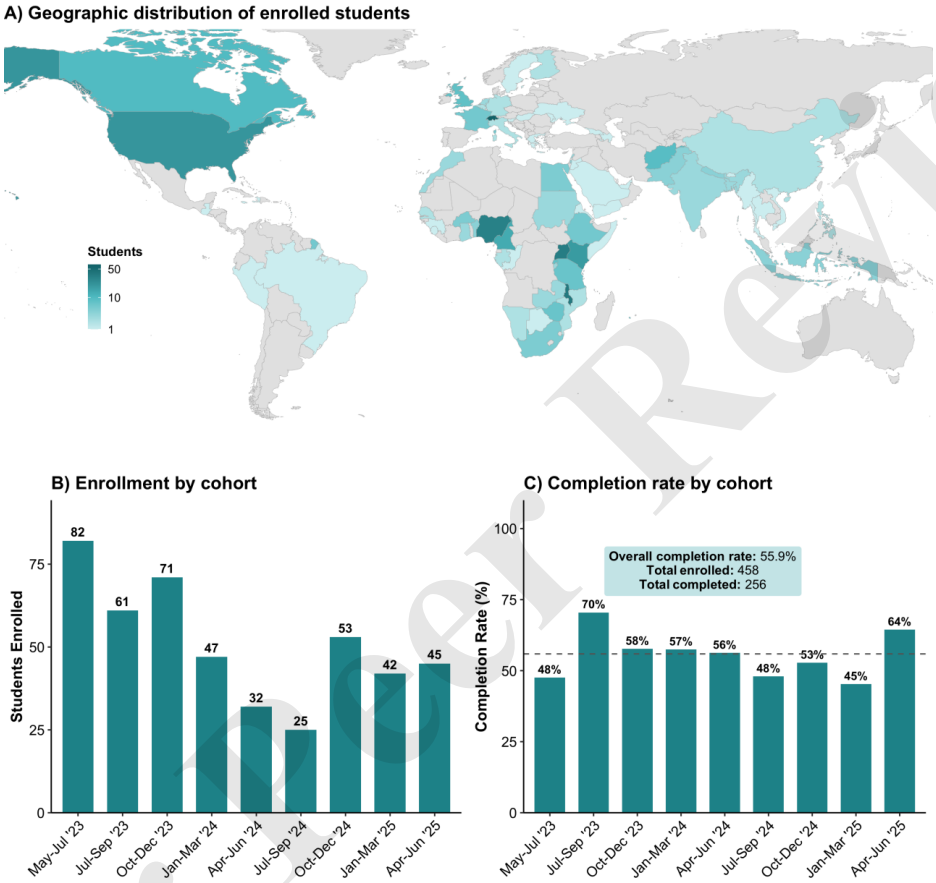

Figure 1: Geographic distribution and enrollment patterns. A) World map showing student distribution by country of residence. B) Number of students enrolled per cohort. C) Course completion rates by cohort, with overall rate indicated.

Predictors of Completion

Logistic regression with multiple imputation was used to explore associations between predictor variables and course completion among 458 students (Figure 2). Of these, 435 had complete data on all predictors; 23 (5%) were missing at least one predictor and were handled via multiple imputation. Neither student status nor employment status was significantly associated with completion: students had similar completion rates to

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non-students (54.8% vs 57.1%; OR = 0.93, 95% CI 0.63-1.35), and employed participants had similar rates to those not employed (55.5% vs 58.2%; OR = 0.9, 95% CI 0.58-1.39). Region of residence was also not significantly associated with completion.

Funding type was associated with completion. Compared to self-funded students (62.7% completion), those receiving full scholarships had significantly lower completion rates (46%; OR = 0.51, 95% CI 0.31-0.83). Partial scholarship recipients had intermediate completion rates (55.9%; OR = 0.75, 95% CI 0.49-1.16). Results were similar in multivariable models and in a complete case sensitivity analysis.

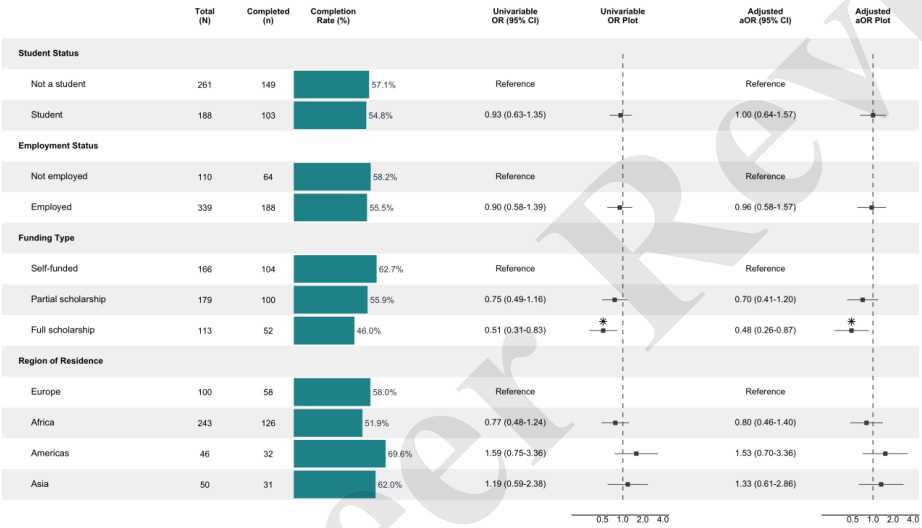

Figure 2: Predictors of course completion. Forest plot showing completion rates and odds ratios (univariable and adjusted) for student status, employment status, funding type, and region of residence. Adjusted models control for cohort. Asterisks indicate statistically significant associations (95% CI excluding 1.0). Odds ratios are pooled estimates from multiply imputed datasets using Rubin's rules.

Characteristics of Survey Respondents

The post-course survey was completed by 206 students, representing a response rate of 45% relative to total enrollment. Survey respondents form the basis for all subsequent analyses of participant characteristics and course experience.

Demographic and educational characteristics of survey respondents are presented in Table 1. Gender was relatively balanced, with 56.3% men (n=116) and 42.7% women (n=88). The most common age group was 31-40 years (41.7%, n=86). The majority (80.1%, n=165) held at least a Master's degree, and most respondents (66%, n=136) were from Africa. Regarding course fees, 61.7% (n=127) received financial aid, 15%

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(n=31) had fees covered by their institution, and 23.3% (n=48) paid the full fee. Most respondents (64.6%, n=133) were non-native English speakers.

Nearly half (43.2%, n=89) had no prior programming experience, but the majority (90.8%, n=187) had some prior statistics experience.

Among survey respondents, 74.8% (n=154) reported completing the course. This is higher than the overall roster completion rate of 55.9%, suggesting that survey respondents over-represent course completers.

Table 1: Characteristics of Survey Respondents (N = 206)

Characteristic	N	Percent
Total	206	-
Gender		
Man	116	56.3
Woman	88	42.7
Missing gender	2	1.0
Age Group (years)		
21-30	69	33.5
31-40	86	41.7
41-50	40	19.4
Above 50	11	5.3
Education Level		
Bachelor's or less	41	19.9
Master's/Postgrad	136	66.0
Doctorate/Medical	29	14.1
Region		
Africa	136	66.0
Asia	17	8.3
Europe	37	18.0
Americas	16	7.8
Financial Aid Status		
Paid full fee	48	23.3

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Characteristic	N	Percent
Received aid	127	61.7
Institution paid	31	15.0
Native English Speaker		
No	133	64.6
Yes	73	35.4
Prior Programming Experience		
None	89	43.2
Some/Significant	117	56.8
Prior Statistics Experience		
None	19	9.2
Some/Significant	187	90.8
Course Completion		
Completed	154	74.8
Did not complete	52	25.2

N is the number of respondents in each category. Percentages are calculated within each characteristic group.

Course Satisfaction and Component Ratings

Among survey respondents, overall satisfaction with the course was high. When asked how likely they were to recommend the course to a colleague (on a scale of 1-10), 96% (n=197) gave a score of 9 or 10 (Figure 3A).

The highest-rated components were pre-recorded video lessons (mean = 9.3, SD = 1.2) and workshop facilitator interaction (mean = 9.2, SD = 1.5) (Figure 3B). The different self-paced quiz types and the use of health datasets also received high ratings (means ranging from 9.1 to 9.2). The online help forum received the lowest ratings (mean = 7.8, SD = 2.4).

To examine whether satisfaction varied by completion status, we compared ratings between the 74.8% of survey respondents who reported completing the course (n=154) and the 25.2% who did not (n=52). Satisfaction was high in both groups: among completers, the mean likelihood-to-recommend score was 9.8 (SD = 0.6) with 95.5% giving scores of 9-10; among non-completers, the mean score was 9.8 (SD = 0.6) with 96.2% giving scores of 9-10. Mean course component ratings were also similar between groups (completers: 9.0, SD = 1.0; non-completers: 8.3, SD = 1.7).

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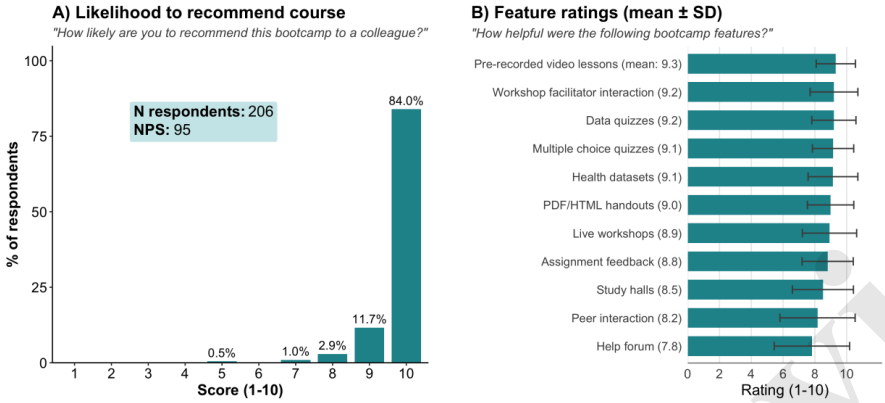

Figure 3: Course satisfaction and component ratings. A) Likelihood-to-recommend score distribution showing the percentage of responses to the question "How likely are you to recommend this bootcamp to a colleague?" on a scale of 1-10. B) Mean course component ratings on a scale of 1-10 (error bars represent ±1 SD).

Course Highlights

Open-ended survey responses provided insight into what students valued most about the course (Figure 4). Live workshops were frequently cited as a highlight (17%, n=35), with respondents emphasizing two main benefits: instructor support and peer collaboration. Instructor support was the most frequently mentioned theme overall (29.6%, n=61): *"The teachers are very flexible, they don't rush you or dismiss you if you're not up to scratch."* Several respondents also valued learning alongside peers during workshops: *"I really appreciated the collaborative atmosphere with classmates which made the course enjoyable and learning smoother."*

Among specific skills, data visualization (18%, n=37) and data wrangling (10.7%, n=22) emerged as particularly valued learning outcomes.

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Course highlights identified by participants

Question: "Please share what you felt were the highlights of the course"

Theme	% of respondents (n)	Exemplar quote from respondent
Support from teaching instructors	29.6% (n=61)	"The teachers are very flexible, they don't rush you or dismiss you if you're not up to scratch. They are very patient. I really appreciate that."
Learning data visualization	18.0% (n=37)	"I especially enjoyed working with packages like ggplot2, which made data visualization intuitive."
Live workshops	17.0% (n=35)	"The workshops were the biggest highlights of the course. Interacting with peers and writing out codes together in virtual classrooms were incredible."
Self-paced prework videos or code-along scripts	13.1% (n=27)	"The course materials: including videos and written tutorials were well curated and up-to-date and served as a valuable references even beyond the course duration."
Course structure	11.7% (n=24)	"The setup of the course worked really well with my learning style. I appreciated the recorded lessons so that I could watch them at my pace. Also, the weekly workshops were very helpful to be able to put the lessons into practice."
Assignment and practice projects	10.7% (n=22)	"For me, the highlights of the course were the assignments, as they allowed me to directly apply what I was learning in practice"
Learning data wrangling/manipulation	10.7% (n=22)	"I found you really focused on the most important R functions that are needed when you want to analyse a dataset : importation, data cleaning, data wrangling,"

Figure 4: Course highlights identified by survey respondents. The table shows the top 7 coded themes from open-ended responses, with the percentage of respondents mentioning each theme and an example quote.

Professional Impact

Respondents were asked to describe how the course had impacted or might impact their professional or academic work (Figure 5). The most common impact was application to ongoing research or data projects (42.7%, n=88), with many describing how they had already begun using R in their current work. Many also reported increased confidence with data analysis (36.9%, n=76), often describing a shift from viewing R as inaccessible to feeling capable of using it independently: *"I'm more confident now to participate in data analysis in my current role."* Other commonly cited impacts included improved data visualization and wrangling skills, preparation for graduate studies, and transitioning from other statistical software to R.

For Peer Review

Professional impacts reported by course participants

Question: "Please explain how the course has helped you in your work or studies, or how you hope the course may help in the fut

Theme	% of respondents (n)	Exemplar quote from respondent
Application to ongoing research/data projects	42.7% (n=88)	"The course helped me analyse the research data I had available. I already started doing this side-by-side as I learned the"
Increased confidence with R or data analysis	36.9% (n=76)	"I'm more confident now to participate in data analysis in my current role. I hope to use R in the upcoming research projects I'll be working on."
Improved data visualization ability	25.7% (n=53)	"I have already started transitioning from STATA to R for data visualizations because of the beauty and power of the ggplot2. I really like this feature!"
Improved data wrangling ability	16.5% (n=34)	"I was missing the necessary skills of efficiently preparing real-life data from data collected during my studies into a format that it can be analysed. This course has helped me a lot with this,"
Preparation for further studies	16.0% (n=33)	"I also plan to start a PhD in public health and this course is of great value as data analysis skills are required for most programs."
Increased self-learning ability	9.2% (n=19)	"the course has provided me with the skills to get started and to understand and learn from online resources more effectively."
Applied for or received job offer using R skills	7.8% (n=16)	"I am now able to do some basic analysis using R. Applied for a data analysis role and was shortlisted. For the assessment, I used R to conduct my analysis"

Figure 5: Professional impacts reported by survey respondents. The table shows the top 7 coded themes from open-ended responses about how the course helped in work or studies, with the percentage of respondents mentioning each theme and an example quote.

Challenges Faced

Respondents were asked to describe any challenges they faced during the course. Time-related challenges dominated open-ended responses (Figure 6). General time constraints were cited by 34% (n=70) of respondents, while 30.1% (n=62) specifically mentioned competing priorities from work or studies. As one respondent explained: *"Time allocated for workshops and other one-to-one meetings with facilitators were overlapping with other duties and activities I had."*

Other challenges were less prevalent but still noteworthy. Infrastructure and access barriers affected some respondents: 8.7% (n=18) reported that poor internet connectivity hampered their access to the course. Programming-related challenges were less common, with 3.9% (n=8) mentioning difficulty retaining R syntax and 4.4% (n=9) citing debugging frustrations.

For Peer Review

Challenges faced during the course

Question: "Please share any particular challenges you faced during the course"

Theme	% of respondents (n)	Exemplar quote from respondent
Time constraints	34.0% (n=70)	"It was difficult with time management to keep on top of things and be ready for the discussions of the assignments during the weekly online sessions."
Competing priorities	30.1% (n=62)	"Time allocated for workshops and other one-to-one meetings with facilitators were overlapping with other duties and activities i had."
Poor internet or low bandwidth	8.7% (n=18)	"I work in remote areas where internet connectivity is a challenge hence i missed some lessons"
Assignment deadlines too tight	6.3% (n=13)	"Time constraints to be up to meet all the requirements on time"
Course moved too quickly	4.4% (n=9)	"The pace for completion was too fast and I had to sacrifice a lot of my extra time"
Debugging and troubleshooting technical issues	4.4% (n=9)	"When you write codes and R says object or so so and so not found without any suggestions on how to go about rectifying the problems"
Difficulty retaining R syntax	3.9% (n=8)	"When I first started, I struggled to memorize codes and to do data wrangling and create graphs using ggplot."

Figure 6: Challenges faced during the course. The table shows the top 7 coded themes (excluding respondents who reported no challenges), with the percentage of respondents mentioning each theme and an example quote.

Recommendations for Improvement

Finally, respondents were invited to suggest improvements for future course iterations. Approximately 40.3% (n=83) of respondents provided only positive feedback without specific recommendations for change.

Among those who offered actionable suggestions (Figure 7), the most common requests were for follow-up courses (5.8%, n=12) and additional practice opportunities (5.3%, n=11). Others requested more advanced or statistical content: *"What I learnt in the course goes as far as descriptive statistics—I want to be able to do some inferential statistics."*

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Recommendations for improvement (Note: 40% gave only positive feedback)

Question: "What recommendations do you have for improving the course?"

Theme	% of respondents (n)	Exemplar quote from respondent
Create follow-up courses	5.8% (n=12)	"I really enjoyed this introductory course and I wish Graph network had intermediate and advance courses in R I could continue with."
Include more practice projects	5.3% (n=11)	"I would recommend some unguided assignments before the final assignment, so that the learner does not feel overwhelmed when having to do the final assignment."
Add advanced topics	4.9% (n=10)	"Though ambitious, I strongly recommend the integration of AI concepts into the R programming course for health data analysts. This approach would empower students with vital skills to utilize AI algorithms for advanced data analysis,"
Add more statistical content	4.9% (n=10)	"What I learnt in the course go as far as descriptive statistics, I want to be able to do some inferential statistics."
Extend course length	4.9% (n=10)	"Maybe include two weeks break at the middle of the course to give room for catch up."
Faster feedback on assignments	4.4% (n=9)	"providing timely feedback on assignments would improve the learning experience"
Extend assignment deadlines	3.4% (n=7)	"I know the course schedule was relatively good, but additional time especially for the final project would be great"

Figure 7: Recommendations for improvement. The table shows the top 7 actionable recommendations (excluding positive feedback with no specific suggestions), with the percentage of respondents mentioning each theme and an example quote.

Discussion

This study evaluated enrollment, completion, and student experience across nine cohorts of an online R programming bootcamp designed for health professionals. The overall completion rate of 55.9% suggests that cohort-based online R training can achieve meaningful engagement among enrolled health professionals.

Nevertheless, nearly half of enrolled students did not complete within the standard timeframe, a non-trivial proportion that warrants consideration. In the qualitative analysis, time-related challenges emerged as the primary barrier to completion. Time constraints were cited by 34% of respondents, with 30.1% mentioning competing priorities from work or studies. These findings are consistent with literature on the barriers adult learners face in online education[14], and underscore a fundamental tension in cohort-based professional education: while the structured format promotes accountability and completion, it also requires sustained time commitment over weeks or months that may conflict with professional duties. Potential solutions include incorporating mid-course breaks, offering alternative scheduling tracks, or developing shorter intensive formats for time-constrained learners.

To complement the self-reported barriers, we examined roster-level predictors of completion. Exploratory analysis found that funding type was a significant predictor: students receiving full scholarships had lower completion rates (46%) than self-funded students (62.7%), while student status and employment status were not significantly associated with completion. Several mechanisms may explain the association between funding type and completion. Scholarship recipients in this program were selected based on demonstrated financial need, and the resource constraints that qualified them for financial aid may independently affect their ability to dedicate time to coursework.

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Additionally, personal financial investment may increase commitment through loss aversion effects[17]; studies of MOOCs have found substantially higher completion rates among learners who pay for verified certificates[18].

Beyond individual-level factors, completion rates also varied substantially across cohorts (45.2% to 70.5%). This variation may reflect differences in cohort composition, variation in instructional delivery across different teaching teams, or external factors such as timing relative to academic calendars or competing professional demands. The specific drivers remain unclear and merit further investigation.

With 96% of survey respondents giving recommendation scores of 9 or 10 out of 10, the program appears to have met learner expectations. Notably, non-completers who responded to the survey reported similarly high satisfaction (mean likelihood-to-recommend score = 9.8) as completers, suggesting that non-completion was driven primarily by time constraints and competing priorities rather than dissatisfaction with the course experience.

Our findings underscore the importance of both asynchronous content delivery and synchronous interaction in online learning. The two highest-rated course components, pre-recorded video lessons (mean = 9.3) and workshop facilitator interaction (mean = 9.2), represent the asynchronous and synchronous components of the flipped classroom model, respectively. This pattern was echoed in qualitative responses, where both the live workshops and the self-paced prework materials were frequently cited as course highlights. These findings align with evidence that blended approaches combining self-paced materials with live instructor interaction outperform purely asynchronous formats[8,14].

The findings in this study should be interpreted in the context of several limitations. Firstly, this observational study lacks a comparison group, preventing causal conclusions about whether the cohort-based format itself drives completion or satisfaction. In addition, survey respondents over-represent course completers (74.8% completion among respondents vs. 55.9% overall), which may positively bias satisfaction estimates. Another concern is that our analysis relies on self-reported data subject to recall bias, and we also lack longitudinal follow-up to assess whether course completion translates into sustained R use in professional practice. Finally, as authors affiliated with the organization delivering this course, our interpretation may be subject to bias toward favorable findings.

In conclusion, this study describes enrollment, completion, and student experience in an online cohort-based R programming course for health professionals. Over half of students completed the course within the standard timeframe, with high satisfaction among survey respondents. Survey respondents valued both self-paced video lessons and live workshop interactions, while time constraints emerged as the primary barrier to completion. These findings suggest that cohort-based formats can engage health professionals in data science training, though scheduling flexibility may be needed to accommodate learners with demanding work responsibilities. Future research should examine long-term skill retention and use controlled designs to compare cohort-based and self-paced approaches.

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Data Availability Statement

The dataset accompanying this manuscript is openly available on Zenodo at https://doi.org/10.5281/zenodo.19857767. The deposit includes the closed-response survey data, the qualitative coding codebook, the coded open-ended responses (with potentially identifying information redacted), and the aggregated tables underlying Table 1 and Figures 1–7. The course roster (country of residence, enrollment, and completion records) and the un-redacted free-text survey responses are not publicly available because participants did not consent to public release; both can be requested from the corresponding author for legitimate research purposes.

Declaration of Interest

In accordance with Taylor & Francis policy and our ethical obligations as researchers, the authors report the following competing interest: all authors are affiliated with The GRAPH Network, which developed and delivers the R Basics and Beyond bootcamp evaluated in this study. Two authors (J. Venkatachalam-Vaz and S. Rodriguez Velásquez) have received remuneration from The GRAPH Network for course development and instruction; the remaining authors contribute to the program through their academic affiliations without direct payment from The GRAPH Network. The authors have a non-financial interest in the program's reputation. The authors report no other potential financial or non-financial competing interests.

Author Contributions (CRediT)

Kene David Nwosu: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft.
Joy Venkatachalam-Vaz: Conceptualization, Data curation, Project administration, Resources, Writing – review & editing.

For Peer Review

Sabina Rodriguez Velásquez: Conceptualization, Data curation, Project administration, Resources, Writing – review & editing.
Sara Botero Mesa: Project administration, Resources, Writing – review & editing.
Olivia Keiser: Supervision, Funding acquisition, Writing – review & editing.

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